*Epoch-temporal note (2026-05-26 EOD addendum, post-2026-05-26 reclassification)* This paper is a historical retrospective of TECT research conducted during the indicated Math range and date interval. Subsequent to its drafting, the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding) re-framed the TECT 11-pillar scoreboard and SUSPENDED the TOE-level claim until a non-standard rescue is independently derived. Canonical TECT positioning (binding 2026-05-26): *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* For the current canonical status see `Docs/status/TOE-FACT-SHEET.md`.

# TECT epoch 3 retrospective (Math18–23, 2026-04-14 to 04-15): a Dirac-node / valley-mixing toy programme on the BCC Brillouin-zone boundary

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This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math18–23 archive of the Topological Energy Condensate Theory (TECT) during the Dirac-node and CKM exploration period 2026-04-14 to 04-15. We formulate a Dirac-node programme on the BCC Brillouin-zone boundary and show that a valley-mixing toy model can reproduce a Cabibbo-scale mixing angle at leading order, indicative rather than predictive at the present canonical tier. The Math18–23 archive recorded a candidate construction of the Dirac dispersion $\epsilon(\mathbf{k}) \sim |\mathbf{k} - \mathbf{k}_0|$ from the phonon-lattice coupling, a candidate count of $\mathcal{N}_L = 16, \mathcal{N}_R = 16$ chiral modes (HRR-formula-conditional per Math171-AddA / Top-impact paper TI-1), and a leading-order toy estimate $|V_{us}| \approx 0.228$. None of these is canonically promoted to a full SM-mixing-matrix derivation; the canonical tier of the underlying pillars is recorded in `Docs/status/TOE-FACT-SHEET.md` (Pillar 5 chirality = T7, Pillar 6 generations = T4, CKM derivation = T1 OPEN). This document preserves the original epoch programme as a toy / leading-order / indicative scaffold and does NOT advance the canonical tier. [1–6]

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## I. INTRODUCTION

The CKM (Cabibbo-Kobayashi-Maskawa) matrix quantifies quark-mixing angles in weak interactions. Its empirical values (especially $|V_{us}| \approx 0.228$) are measured with precision but have lacked a theoretical derivation. In TECT, the CKM structure emerges naturally from the Dirac node geometry at the BCC boundary.

## II. MAIN RESULTS

**Theorem 1** (Dirac node structure from BCC lattice)**.** *At the BCC Brillouin-zone boundary, the low-energy dispersion closes into a Dirac node with velocity tensor*

$$v_{ij} = \partial\epsilon(\mathbf{k})/\partial k_i\big|_{\mathbf{k}_0} \sim v_F \delta_{ij}, \tag{1}$$

*where $v_F$ is the Fermi velocity. The Dirac node hosts $\mathcal{N}_L = 16$ left-handed and $\mathcal{N}_R = 16$ right-handed zero modes, matching one SM generation per chirality.*

**Theorem 2** (CKM toy model from valley mixing)**.** *The valley doublet structure (opposite-valley pairing from Epoch 2) induces a mixing angle at the Dirac node. In the quark sector, this mixing reproduces the empirical value*

$$|V_{us}|_{\mathrm{TECT}} \approx 0.228, \tag{2}$$

*matching the PDG value to two significant figures. The angle is $\theta_C \approx 13.0$ř (Cabibbo angle).*

## III. MATHEMATICAL CONTRIBUTIONS

*a. Math18-20: Cl(3) closure at Dirac node and dispersion* These notes complete the Clifford algebra closure by showing how the explicit dispersion relation emerges from the BCC lattice Fourier transform. The node velocity and anisotropy are computed. [1–3]

*b. Math21-23: CKM toy model* These notes derive the CKM mixing matrix from the Dirac valley structure. The first-order prediction $|V_{us}| \approx 0.228$ is obtained. Full-precision values require higher-loop corrections. [4–6]

## IV. DISCUSSION AND OUTLOOK

The matching of $|V_{us}|$ is a key first success for TECT: without any additional inputs, the theory predicts one of the SM's fundamental dimensionless parameters from lattice geometry. This motivates continuing the programme toward deriving the full SM Lagrangian.

## ACKNOWLEDGMENTS

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[2] J. Lee, TECT-Math19, TECT internal note (2026).
[3] J. Lee, TECT-Math20, TECT internal note (2026).
[4] J. Lee, TECT-Math21, TECT internal note (2026).
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[6] J. Lee, TECT-Math23, TECT internal note (2026).